

Bridge Day 16 STUDENT QUIZ

Exam 2 Rehearsal: Loops, Lists, Mutation, and Filtering

Learning sequence: Predict - Trace - Explain - Test - Interpret

Monday, July 27, 2026 • 20 points • target 17 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, nested condition tracing, break/continue, indexed mutation, compound list filtering

Scaffold level: minimal-scaffold Exam 2 rehearsal

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.25, 18.26, 18.27.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace skip, running state, collection, and early stop. - 6 points

```
values = [3, -2, 5, 0]
result = []
running = 0

for index in range(len(values)):
    if values[index] < 0:
        continue
    running = running + values[index]
    if running > 6:
        result.append(running)
        break
    result.append(values[index] * 2)

print(result, running, index)
```

Question: Give a complete reached-pass ledger, including skipped statements and the exact final output.

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Item 2. Diagnose final-list evidence that disagrees with the total. - 4 points

```
values = [4, -3, 7, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Question: Give actual output, explain the mismatch, and write an indexed repair that makes the list and total describe the same final state.

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Complete program - 10 points

- Read threshold as an integer and use values = [12, -4, 7, 20, 7].
- Loop by index and replace negative values with 0 before any summary.
- Append a final value to selected only when it is at least threshold and even.
- Compute the total of all final values and the index of the first largest final value without max or index.
- Print Final, Selected, Total, and Largest index on separate labeled lines. Give a threshold-equality test.

Program workspace**Test input, expected result, and why this test matters**